GuardSynth: Checkable Behavioral Constraints for CoC-Augmented Candidate Selection

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Abstract

Natural-language reasoning supports autonomous-driving decisions in vision–language and vision–language–action models. Explanations alone leave open whether models will follow newly supplied constraints and select preferred admissible actions. Interactions among constraints motivate checking their construction; we evaluate single-constraint learning tasks and a separate bounded composition diagnostic. GuardSynth binds constraint fields to observations, supplied rules, and modeling assumptions in Evidence-Bound Lifecycle Contracts (EBLC). It checks satisfiability and candidate admissibility within supported models, generates Controlled Natural Language (CNL), and appends it to Chain-of-Causation (CoC) reasoning. Supported parsers check specification–sentence correspondence. The preferred-selection score is the percentage of predictions selecting the highest-scoring candidate satisfying all applicable constraints under the supplied synthetic objective. Rule-pair accuracy requires both rule-paired reference labels. In synthetic candidate-selection tasks, Qwen3-VL-2B-Instruct adapters receive identical evaluation CNL with matched initialization across

three seeds. Under cross-entropy alone, CNL-trained adapters achieve preferred-selection scores of 93.49–98.47%, constraint violation rates of 0.00–1.56%, and rule-pair accuracy of at least 86.98%; CoC-only-trained adapters fine-tuned without rule text have constraint violation rates of 6.25–17.92% and reach at most 56.77% rule-pair accuracy. Post-training fault injection detects all 288 supported faults and accepts all 48 valid cases, but misses 48 jointly wrong evidence–specification controls. Document-grounded composition checks find 58 incompatible configurations among 176 authored configurations, including 12 undetected by pairwise checks. GuardSynth provides checkable constraint construction and evidence that training with constraints improves rule-dependent selection relative to rule-free fine-tuning.

Keywords: Autonomous driving, behavioral constraints, Chain of Causation, controlled natural language, specification checking, vision-language models

1 Introduction

Vision-language models (VLMs) and vision-language-action (VLA) models connect visual observations and language information to action selection or trajectory prediction, with autonomous driving as a prominent application. Recent work uses supervised fine-tuning and reinforcement learning to improve language reasoning and action prediction together [29]. Other proposals integrate future-scene modeling into planning or maintain language-based memories of past observations to support decisions over time [41, 46]. Evaluation also extends from individual prediction performance to closed-loop driving and behavioral changes under external instructions [20, 47].

Despite these advances, reliably connecting a model's judgment to the required behavior remains difficult. Drive the Thoughts reports only 74% agreement between predicted trajectories and reasoning traces judged reliable [48]. Language2Drive reports that models can succeed under one instruction yet fail under a feasible alternative when the initial scene is held fixed [47]. Generating an explanation or driving successfully under one condition therefore does not establish the ability to adjust behavior when external conditions change. We focus on using externally supplied behavioral rules as conditions for action selection. A supplied behavioral rule states a condition to apply in a particular case; its source may be a traffic regulation, an operator instruction, or an experimental policy. The same scene and action candidates can admit different actions under different rules.

Using such rules is distinct from explaining the reason for a driving decision. Driving models that reason in language express how an observed situation leads to a decision. Alpamayo-R1's Chain of Causation (CoC) is one example: it learns causal relationships between observations and decisions together with trajectory prediction [29]. Such reasoning need not fully specify an external rule's applicability conditions and numerical parameters as a case-specific behavioral constraint. Here, a behavioral constraint makes a supplied rule concrete through applicability conditions, action restrictions, timing, and release conditions, so that action admissibility can

be assessed. The scene and causal explanation can remain valid while different speed bounds, waiting durations, or stop-release conditions change the admissible actions. An action consistent with its explanation can therefore violate a separately supplied constraint.

Behavioral constraints determine which actions are admissible, but do not by themselves rank the admissible actions. For example, always choosing the longer wait under two different waiting rules may satisfy both rules while missing an opportunity for earlier permitted progress. Constraint compliance and preferences among admissible actions must therefore be distinguished. In our candidate-selection task, the goal is to satisfy the applicable constraints and select the admissible candidate with the highest supplied progress score. This score is a synthetic value assigned by the task generator; it is not a direct measurement of distance traveled or long-term driving efficiency.

Learning this selection requires training examples to make the relationship among scenes, applicable rules, and actions explicit. It also requires checking whether constraints assembled from observations, supplied rules, and modeling assumptions express the intended behavioral conditions, and assessing whether the model uses them in its choices. Including a rule in the input alone does not establish such use. Our preliminary Alpamayo-R1 diagnostics, without additional training, produced predictions violating explicitly supplied speed and position-hold rules on sampled PhysicalAI [30] and converted nuScenes [4] inputs. In a separate controlled candidate-selection study, we therefore test whether including constraints in training improves constraint compliance and objective-based candidate selection when every model receives the same rule at evaluation.

Behavioral constraints supplied for learning require rigorous validation to determine whether their combination satisfies the specified safety requirements under the stated assumptions. Combining behavioral conditions can restrict a necessary safety response. The 2018 fatal Uber collision in Tempe illustrates this concern. The National Transportation Safety Board (NTSB) reported emergency recognition 1.2 s before impact, followed by 1 s of braking suppression intended to reduce false-alarm responses. Only about 0.2 s remained afterward. The system also lacked maximum allowed braking solely for mitigation when collision avoidance exceeded its braking limits. The NTSB identified increased safety risk from this design [28]. This illustrates why suppression timing must be checked against collision-response requirements.

In the 2023 Cruise pedestrian collision, incorrect lateral-collision classification led to a pullover after an initial stop, dragging the pedestrian [9]. This case highlights the need to examine the conditions governing transitions between remaining stationary and moving. Rule 129 of the United Kingdom's Highway Code generally prohibits crossing double white centre lines when the line on the driver's side is solid, but allows crossing when necessary to pass a stationary vehicle, provided the road is clear [10]. This rule illustrates why specifications must preserve exceptions in the original rules: a constraint that omits this exception could block a lawful manoeuvre and prolong unnecessary waiting. Our learning experiments use single constraints, while a separate logical diagnostic combines conditions drawn from these and related documents, including the Tempe braking suppression and a restricted encoding of the Rule 129 exception, under explicitly authored states. It examines the joint feasibility

of the stated conditions, rather than reconstructing the accidents or interpreting the regulations in full.

These examples motivate treating behavioral constraints as checkable objects with explicit evidence and applicability. Supplying them as natural-language learning inputs raises three problems. First, the evidence and roles of observed facts, supplied rules, and modeling assumptions must be explicit, so that observation errors, rule misapplication, and unsuitable assumptions can be examined separately. Second, the specification must express the intended action restrictions and release conditions. For the learning specifications evaluated here, satisfiability and candidate admissibility must be checked under the supplied inputs and assumptions. When several supplied conditions apply, joint satisfiability must also be assessed under an explicit shared context. Third, mechanical checking requires a formal representation, whereas the model learns from natural language. Meaning must be preserved when the specification is rendered as a sentence, and model behavior must be evaluated separately: correspondence between a sentence and its specification does not establish constraint compliance.

To address these problems, we propose GuardSynth. It first binds each constraint field to an observation, a supplied rule, or a modeling assumption, separating their roles. It represents applicability, action restrictions, persistence and release, timing, and evidence in Evidence-Bound Lifecycle Contracts (EBLC). Supported specifications are lowered to Satisfiability Modulo Theories (SMT) queries for satisfiability and candidate-admissibility checks within a restricted logical model. A separate composition diagnostic checks source conditions jointly under supplied contexts. GuardSynth then generates Controlled Natural Language (CNL) from the specification and appends it to the original CoC. For supported fixed grammars, parsing the generated sentence recovers fields for comparison with the specification. Our evaluation concerns the supported specifications and synthetic candidate-selection tasks used in this process.

The evaluation separates the behavioral effect of training with constraints from specification checkability. Constraint violation rate is the proportion of all predictions that select a prohibited candidate. The *preferred-selection score* is the proportion that select the most preferred admissible candidate under the specified objective. In our tasks, the preferred candidate has the highest supplied progress score among admissible candidates; selecting its label receives a score of one, and every other output receives zero. The mean is reported as a percentage. Rule-pair accuracy is the proportion of pairs for which both reference labels are selected when the scene and candidates are fixed and only the rule changes. These metrics distinguish constraint satisfaction from objective-based selection within the supplied candidate set.

We evaluate fresh Qwen3-VL-2B-Instruct adapters [32] with matched initialization and the same CNL at evaluation over three training seeds. Training with CNL under cross-entropy loss alone yields three-seed mean constraint violation rates of at most 1.56% and rule-pair accuracy of at least 86.98% across pedestrian entry and the two position-hold representations. This controlled comparison supports including case-specific constraints in training data as well as supplying them at inference for the evaluated adapters. Across six behavioral tasks, both natural-language guards (P1)

and EBLC-derived CNL (P2) improve selection of the preferred candidate under the supplied rule; this comparison changes rule availability as well as training inputs.

A separate fault-injection evaluation of the two specifications used to generate learning inputs detects all 288 errors of six supported types and accepts all 48 valid cases. All 48 jointly wrong evidence–specification controls remain undetected: the evidence record and specification are changed to the same wrong value. This identifies a boundary between internal correspondence checks and independent confirmation of the evidence itself. Evidence correspondence, sentence correspondence, and boundary probes for logic–rule comparison were added after training for this evaluation; their results are interpreted separately from the causes of behavioral improvement.

A separate composition diagnostic combines manually formalized conditions from official documents with explicit modeling assumptions and authored states, candidates, or test objectives. Among 176 configurations, it identifies 58 incompatible combinations, including 12 three-way conflicts for which every pair is satisfiable. All 118 compatible combinations are retained. These results concern joint logical feasibility under explicit contexts, rather than VLM behavior or a general vehicle-safety guarantee.

Our contributions are:

1. A procedure for constructing case-specific behavioral constraints tied to their evidence. It separates observations, supplied rules, and modeling assumptions and links them to EBLC specifications with applicability, action restrictions, persistence, and release conditions, making the components and evidence of learning constraints explicit objects of checking.
2. The GuardSynth development path connecting formal specification checks to natural-language learning inputs. It lowers supported specifications to SMT for satisfiability and candidate-admissibility checks within a restricted model, appends specification-derived CNL to the original CoC, and checks specification–sentence correspondence by parsing supported grammars. The same Core/SMT compiler checks bounded source combinations under shared contexts in a separate diagnostic that does not produce learning inputs.
3. An experimental evaluation separating the effect of training with constraints from specification checkability. Matched initialization and evaluation inputs test improvements in constraint compliance and objective-based candidate selection, while separate fault injection identifies complementary checking layers and their detection limits. A separate composition diagnostic compares individual, pairwise, and joint consistency checks for supplied sources. Model behavior and the correctness of learning-constraint construction are assessed separately.

Section 2 reviews related work, and Section 3 presents the problem setting and GuardSynth. Sections 4 and 5 describe the experimental setup and results. Section 6 discusses implications and limitations, and Section 7 concludes.

2 Related Work

We distinguish three questions: how a model uses a supplied behavioral rule, how that rule is expressed and checked as a constraint, and how the constraint is delivered as learning input. The following studies address different parts of this path; constraint compliance and preferences among admissible actions are separate assessment targets.

2.1 Reasoning-Based Driving and Reasoning Faithfulness

Recent driving models connect visual observations, language reasoning, and action planning. Alpamayo-R1 combines Chain of Causation (CoC) reasoning with trajectory prediction and uses supervised fine-tuning and reinforcement learning to improve reasoning–action alignment [29]. DriveLM connects perception, prediction, and planning through graph-structured question–answer pairs, while DriveVLM links scene description and analysis to hierarchical planning [36, 37]. Drive-R1 addresses reasoning–planning misalignment and history-based shortcut learning through reward design [25]. These studies develop learnable representations of driving decisions.

Reasoning faithfulness and reasoning–action consistency raise related but distinct questions. Turpin et al. [38] show that chain-of-thought explanations can omit factors influencing an answer, while Lanham et al. [24] investigate whether predictions depend on the stated reasoning. In driving, Drive the Thoughts checks whether predicted trajectories match commitments expressed in VLA reasoning [48]. These assessments concern reasoning faithfulness or reasoning–action consistency; constraint compliance with externally supplied behavioral constraints is a separate assessment target.

2.2 Traffic-Rule Formalization and Safety Enforcement

Formalized traffic rules provide a foundation for expressing admissible driving behavior. Esterle et al. [11] encode traffic rules in linear temporal logic (LTL), and Maierhofer et al. [26] formalize interstate rules for compliance assessment. Rulebooks represent rule priorities and the ordering of behavioral outcomes, whereas Responsibility-Sensitive Safety (RSS) defines mathematical conditions for safe behavior and responsibility [6, 35].

Checking the formalized rules themselves is also an established concern. Wan et al. [39] combine equivalence verification and model checking to assess the semantic consistency and correctness of digital traffic rules. DriveReg instead retrieves applicable regulations and incorporates them into language-model reasoning, supplying rules at inference [5]. One of our controls shares this principle: adapters trained without rule text receive the rule only at evaluation (Table 7).

Safety enforcement acts at another interface. Shielding restricts or corrects actions to satisfy temporal-logic specifications during learning or execution [1]. Context-conditioned Hamilton–Jacobi reachability incorporates geometry, friction, and disturbances into a safety representation used by a control filter [34]. GuardSynth acts on training data and leaves all candidates available at inference, making it complementary to mechanisms that filter proposed actions.

2.3 Natural-Language Formalization and Controlled Natural Language

Natural-language formalization connects readable requirements with machine-interpretable specifications. NL2TL abstracts atomic propositions and learns to translate requirements into temporal logic [7]. nl2spec exposes correspondences between language fragments and subformulas so users can revise ambiguous or erroneous translations [8]. Their central output is a formal specification. GuardSynth's language-generation stage reverses this mapping: it generates CNL from checked specifications and evaluates correspondence by parsing sentences back with a supported fixed grammar to recover their fields (Section 3.5).

Controlled natural language (CNL) restricts language to support a defined interpretation. Attempto Controlled English (ACE) connects restricted English syntax with logical representations [14], while Kuhn [23] classifies CNLs by their properties and purposes.

2.4 Incorporating Constraints into Learning and Generation

Constraints can influence models through several mechanisms. Semantic loss converts logical knowledge into a training objective that encourages outputs to satisfy constraints [45]. We compare a categorical adaptation of this objective with supplying rules as input text (Section 4.4; Eq. 9). Giunchiglia et al. [15] survey a broader range of approaches for combining deep learning with logical constraints. Constrained decoding acts during generation: Grid Beam Search enforces the inclusion of specified words or phrases, while guided generation restricts outputs using regular expressions and grammars [17, 44]. These mechanisms address constraint satisfaction at different levels; a training penalty, a lexical requirement, and a grammatical restriction provide different forms of control.

Training with task instructions or rationale supervision can improve inference-time task performance. Wei et al. [43] instruction-tune FLAN to improve zero-shot performance on unseen task types, and Hsieh et al. [18] use model-generated rationales as auxiliary training targets for smaller models. Our study evaluates this principle with case-specific behavioral constraints and numerical parameters. Rule-paired evaluation holds scenes and candidates fixed while changing the supplied rule, and specification checking assesses construction and correspondence separately from the model's choices.

Natural-language principles offer another route. Constitutional AI uses principles to guide self-critique, revision, and AI-feedback training [2]. Deliberative Alignment trains models to recall and reason over a reusable safety policy; category-specific specifications guide training-data generation and filtering [16]. The policy text is then removed from the prompts used for supervised fine-tuning. GuardSynth retains case-specific constraints and their numerical parameters in both training and evaluation inputs. Our P1 condition adds natural-language guards directly, while P2 uses EBLC-derived CNL at the same learning interface. Their comparison examines constraint wording and construction paths within this shared interface. Our preceding work

showed that natural-language guards improve action selection [21]; GuardSynth adds a checked construction path for such constraints.

2.5 Positioning of This Work

GuardSynth combines these foundations at the interface where behavioral constraints become learning data. It binds constraint fields to observations, supplied rules, or modeling assumptions; represents supported constraints as Evidence-Bound Lifecycle Contracts (EBLC); checks satisfiability and candidate admissibility within restricted logical models; and generates CNL appended to the original CoC. Its contribution is this connected development path. We also test whether constraints supplied during training improve rule-dependent selection relative to adapters fine-tuned without rule text and supplied the same rules only at evaluation. Checks are limited to supported profiles and supplied evidence. Model evaluation distinguishes constraint violations from selection of the preferred admissible candidate under a supplied objective. This path complements learning objectives and runtime filters; its evaluation uses supervised candidate selection rather than a comparison of supervised and reinforcement learning.

Table 1 summarizes the main interfaces and assessment targets of the selected research.

Table 1 Main interfaces and assessment targets of related research

Research direction	Main interface	Main assessment or control target
Reasoning-based driving [25, 29, 36, 37]	Observations, reasoning, and planning	Driving performance and reasoning–action alignment
Reasoning assessment [24, 38, 48]	Explanations, predictions, or trajectories	Reasoning faithfulness or reasoning–trajectory consistency
Traffic-rule formalization and verification [6, 11, 26, 35, 39]	Informal rules and formal behavioral models	Rule semantics, priorities, correctness, or behavioral compliance
Regulation-augmented reasoning [5]	Retrieved regulations and scene reasoning	Selection and use of applicable regulations
Shielding and safety filtering [1, 34]	Proposed actions or control inputs	Satisfaction of specified safety conditions
Natural-language formalization [7, 8]	Language requirements and temporal logic	Translation quality and, in nl2spec, interactive correction
Logical learning and constrained decoding [15, 17, 44, 45]	Training objectives or generation procedures	Logical satisfaction, required lexical content, or output structure
Instruction tuning and rationale distillation [18, 43]	Task instructions or rationale supervision	Instruction generalization or data-efficient distillation
Natural-language principle learning [2, 16]	Principles, training examples, and policy reasoning	Alignment with supplied principles or safety specifications
GuardSynth	**Source-bound contracts and CoC learning inputs**	**Supported contract properties, field–CNL correspondence, and separately measured constraint compliance and preferred selection**

3 Proposed Method

GuardSynth makes the evidence and meaning of behavioral constraints explicit, checks them within supported logical models, and expresses them as language for learning. It retains the original CoC as a causal explanation of the driving decision and appends a separately constructed constraint on admissible actions. The three problems identified in the Introduction organize the method: construction links fields to evidence, specification checking assesses the supported logical meaning, and language generation connects that meaning to learning inputs. Pedestrian entry provides the running example; position holding extends the admissibility condition to sampled positions over time.

3.1 Problem Setting

Each example contains a scene input, the original CoC, a set of action candidates $\mathcal{A}$, and a supplied behavioral rule. A rule is an externally provided condition; a behavioral constraint makes its applicability, restriction, and release precise enough to assess admissibility. Supplied rules may represent experimental policies or operating instructions, so constraint compliance is distinguished from compliance with traffic laws and regulations. Let h_t denote the observations and state history supplied up to decision step t. A behavioral constraint C defines the admissible candidate set

$$A_C(h_t) = \{a \in \mathcal{A} \mid a \text{ satisfies } C \text{ under } h_t\}. \tag{1}$$

When several constraints apply, their admissible sets are intersected. Admissibility here means satisfaction of the stated constraints; physical feasibility requires the relevant vehicle and environment dynamics to be modeled as well.

Constraints determine admissibility, whereas the selection objective ranks admissible candidates. We use the admissible candidate with the greatest progress score $J(a)$ as the preferred candidate a^*, whose label is the reference label used for training and evaluation:

$$a^* = \underset{a \in A_C(h_t)}{\arg\max}\ J(a). \tag{2}$$

The task generator supplies the synthetic progress score $J(a)$; it is not part of the logical constraint or an SMT optimization objective. It ranks candidates for this task rather than measuring distance traveled or long-term driving efficiency. An admissible choice with a lower score therefore differs from the preferred candidate without violating the constraint. The preferred-selection score in Section 4.2 measures the frequency of selecting a^*, rather than the magnitude of a quality loss.

The learning model outputs a label for one of the supplied candidates. Even when candidate descriptions contain time-indexed coordinates, the model does not generate a new trajectory. This candidate-selection task differs from the Alpamayo trajectory-generation diagnostics reported in the Introduction.

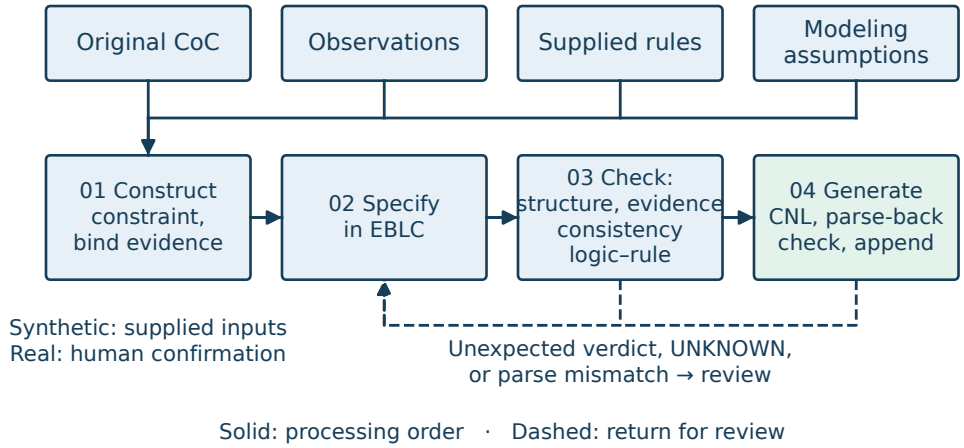


Fig. 1 GuardSynth workflow. Four inputs feed four stages. Unexpected solver verdicts, UNKNOWN, or a sentence-parsing mismatch return the specification for review. Synthetic experiments use supplied inputs; real scenes require human confirmation of applicability. In the reported experiments, evidence and sentence correspondence were assessed after training (Section 3.5).

3.2 Workflow

GuardSynth takes the original CoC, scene observations, supplied rules, and modeling assumptions through four stages (Fig. 1).

1. **Constraint construction:** identify the behavioral restriction and associate its required information with observations, rules, or assumptions.
2. **Specification:** represent applicability, restricted actions, persistence, release, and evidence in an EBLC specification.
3. **Checking:** inspect the structure and logical interpretation and compare them with supplied rules and evidence records where supported.
4. **Language generation:** render the specification as CNL, assess sentence correspondence where a supported parser is available, and append the sentences to CoC.

The outputs are an EBLC specification, check records, and sentences for the learning input. The aim is to maintain correspondence among specification meaning, check outcomes, and language. Unexpected or UNKNOWN solver outcomes require review. Structural validation is the renderer's implemented precondition; logical check outcomes are assessed separately and retained for review.

3.3 Constraint Construction and Evidence Binding

CoC combines observations with a causal explanation; a behavioral constraint is a separate element that determines which actions are permitted under specified conditions. A selection objective expresses preferences among permitted actions.

Table 2 Roles of information used to construct a constraint

Role	Example	Use in interpretation
Observation	Time at which the pedestrian leaves the conflict zone	Supplies environmental state and event time
Supplied rule	Wait at least 0.5 s after clearance	Determines the temporal boundary of admissibility
Modeling assumption	No reoccupation after clearance	Defines the scope of the temporal interpretation

Constraint construction identifies the applicable situation, the restricted action, and its bound or release condition. The required information is associated with one of the three roles in Table 2.

This separation distinguishes observed facts from required behavior. Clearance time can be observed, but the required wait is determined by a separate rule. Waiting durations and permitted distances are therefore taken from supplied rules rather than inferred from video or CoC wording.

Conditions and values are linked to their supporting records, as illustrated in Table 3. If a constraint is wrong, these links identify the observation, rule, or assumption requiring review. Recording a source does not establish that the source is correct.

In the synthetic experiments, the generator supplies observations, rules, and candidate measurements to predefined specification constructors. The original natural-language CoC is retained; these experiments do not parse its unrestricted meaning into EBLC. Real-scene use requires human confirmation of the relevant entities, regions, event times, and rule applicability. Automatic extraction of this information from video and its quality are not evaluated in this study.

3.4 EBLC Specifications

An Evidence-Bound Lifecycle Contract (EBLC) records a behavioral constraint through its subject and target, applicability, action restriction, persistence and release, time and units, and supporting evidence and assumptions. "Evidence-bound" denotes the connection between values and their source records. "Lifecycle" covers activation, persistence, release, and, where needed, reactivation. Each task uses the required elements; only combinations supported by the validator and logical interpretation are accepted.

Pedestrian-entry constraint. This constraint prohibits entry while the conflict zone is occupied and requires a specified wait after clearance. Table 3 shows the semantic contents of its specification for a 0.5 s rule and of the position-hold specification described below, without implementation identifiers, together with the CNL phrase generated from each element.

Clearance time is an observation supplied during checking, not a value stored in this specification. Let τ be clearance time, u the candidate's entry time, Δ the

Table 3 EBLC specifications and the CNL phrase generated from each element

Element	Value	Role	Generated phrase
Pedestrian entry, 0.5 s wait			
Subject and target	Ego vehicle; pedestrian conflict zone	Restriction scope	"the pedestrian conflict zone"
Restricted action	No entry during occupancy	Action restriction	"Do not enter ... while it is occupied."
Release requirement	At least 0.5 s of clearance	Supplied rule	"Enter only after the zone has remained clear for at least 0.5 s."
Assumption	A single clearance transition with no reoccupation	Interpretation scope	Not rendered
Evidence	Supplied synthetic rule and its source-record identifiers	Traceability	Not rendered
Position hold, 2.0 s			
Subject and target	Ego vehicle; initial XY position	Restriction scope	"the ego vehicle's initial position"
Hold duration	2.0 s	Supplied rule	"Hold ... for at least 2.0 s."
Checked samples	Listed waypoints with $t \leq 2.0$ s, endpoint included	Interpretation scope	"At every listed waypoint with t <= 2.0 s, including the endpoint,"
Restricted action	Radial distance at most 0.2 m	Action restriction	"the radial XY distance from the initial position must be at most 0.200 m."
Release	Motion permitted after the hold	Release	"Movement after that period is permitted."
Assumption	Waypoints relative to the initial origin	Interpretation scope	Stated in candidate descriptions
Evidence	Supplied synthetic rule and its source-record identifiers	Traceability	Not rendered

required wait, and e its Boolean entry flag. Under the no-reoccupation assumption, admissibility requires

$$e \Rightarrow u \geq \tau + \Delta. \tag{3}$$

The specification supplies Δ, whereas the environment and candidate supply τ, u, and e. This separation permits different rules to be applied to the same environment and different candidates to be checked against the same specification.

For example, clearance at 11.0 s permits entry from 11.5 s under a 0.5 s rule and from 12.5 s under a 1.5 s rule. The scene and candidates can be identical while the admissible set changes. A non-entry candidate satisfies both rules; its selection depends on progress.

Position-hold constraint. This constraint requires a candidate to stay within a permitted radius of its initial position for a specified duration. Let T be the hold duration, R the permitted radius, and (x_i, y_i) the candidate's sampled coordinates relative to its initial position. Admissibility requires

$$\bigwedge_{t_i \leq T} \left(x_i^2 + y_i^2 \leq R^2\right). \tag{4}$$

We use $R = 0.2\,\mathrm{m}$ and $T \in \{0.5, 2.0\}\,\mathrm{s}$. The interval includes its endpoint; movement after the hold period is unrestricted. Only the listed samples are checked, rather than continuous motion between samples or vehicle dynamics.

The same candidates are presented in two forms. **Time-indexed XY coordinates** require the model to identify samples within the hold interval and assess their distances from the initial position. **Interval maxima** supply every candidate with both $D_{\max}(0.5\,\mathrm{s})$ and $D_{\max}(2.0\,\mathrm{s})$, where $D_{\max}(T) = \max_{t_i \le T} \sqrt{x_i^2 + y_i^2}$, regardless of the active rule; the model must still select the required interval and compare its maximum with the permitted radius. Reference admissibility is computed from the original integer coordinates and squared distances, not rounded maxima displayed in the prompt. Comparing the two forms helps distinguish coordinate-calculation demands from constraint-based selection, although it also changes the input representation.

Numerical bounds. Speed, clearance, and entry-delay tasks place an upper or lower bound on a candidate measurement. With completion flag q, measurement x, and bound b, the basic form is

$$q \Rightarrow (x \ \mathrm{op} \ b), \qquad \mathrm{op} \in \{\le, \ge\}. \tag{5}$$

Section 3.7 explains the treatment of non-completing candidates and its difference from the generated language.

Activation, persistence, and release. The preceding constraints are evaluated from supplied values. A separate qualitative specification tracks obligations when evidence changes over time: valid activation or release evidence updates the state, while uncertain or conflicting evidence preserves the previous state. Its state-transition checks are reported in Supplementary Section S3; the learning experiments do not use this specification.

Table 4 connects each task to its admissibility condition. The first six tasks form the six-task comparison; position holding is evaluated in both the fresh-adapter and continuation studies. All numerical bounds are supplied experimental rules. Entry delay δ is a duration, whereas u is an instant. In stopping, o is position relative to the stop line, with negative values before the line; a_{peak} and j_{peak} denote peak deceleration and jerk.

3.5 Specification Checking

The admissibility conditions in Section 3.4 are translated into logical formulas. We use SMT and Z3 [3, 27] and distinguish property queries from candidate queries. The learning specifications are evaluated through satisfiability, candidate admissibility, and concrete logic–rule comparisons. Property-query semantics are illustrated below; the separate qualitative lifecycle example in Supplementary Section S3 is not used for learning.

Specification properties. Let Φ combine the specification and its assumptions, and let P be a property of interest. The counterexample query is

$$\Phi \wedge \neg P. \tag{6}$$

Table 4 Task admissibility conditions with representative supplied rules. q denotes completion and e denotes entry.

Task	Admissibility condition
Maximum speed	$q \Rightarrow v \leq 4.4\,\mathrm{m/s}$
Minimum clearance	$q \Rightarrow d \geq 2.0\,\mathrm{m}$
Minimum entry delay	$q \Rightarrow \delta \geq 2.5\,\mathrm{s}$
Pedestrian entry	$e \Rightarrow u \geq \tau + \Delta$
Stopping	$q \Rightarrow (o \leq 0 \wedge a_{\mathrm{peak}} \leq 5.8\,\mathrm{m/s^2} \wedge j_{\mathrm{peak}} \leq 7.4\,\mathrm{m/s^3})$
Lane change	$q \Rightarrow g \geq 5.8\,\mathrm{m}$
Position holding	$\bigwedge_{t_i \leq T} (x_i^2 + y_i^2 \leq R^2)$

UNSAT means that no counterexample exists within the encoded scope. SAT means that an assignment satisfies the specification while violating the property. Because an inconsistent specification implies every property vacuously, the satisfiability of Φ is checked separately.

For the entry constraint, the requirement that entry occur only after the release boundary can be expressed as $P = (e \Rightarrow u \geq \tau + \Delta)$. Its counterexample query, $\Phi_C \wedge e \wedge (u < \tau + \Delta)$, must be UNSAT. Removing the release restriction permits a SAT counterexample. This illustrates property-query semantics; the fault-injection logic–rule comparisons below use concrete boundary assignments.

Candidate queries. Let $B(a)$ fix candidate a's measurements and completion or entry flag. We check

$$\Phi_C \wedge B(a). \tag{7}$$

SAT means that the candidate is compatible with the constraint, and UNSAT means that it is incompatible. This query checks admissibility, not whether the candidate is the preferred one.

Before training, the pedestrian-entry and position-hold specifications used for learning underwent structural validation, satisfiability checks, and candidate checks.

Checking layers. Five checks compare different objects.

1. **Structural check:** verify supported field combinations, values, and units.
2. **Satisfiability check:** confirm that the base specification is satisfiable, so that properties do not hold vacuously.
3. **Logic–rule comparison:** compare logical verdicts for concrete candidates with verdicts computed by independent integer rules from the supplied rule. Archived action candidates are evaluated together with boundary probes: for pedestrian entry, entry or non-entry at $-1, 0, +1$ ms around release; for position holding, distances of 199, 200, and 201 mm, with diagonal norm checks, at every listed

non-initial sample, including samples after release. The 1 ms spacing tests logical boundaries, not vehicle-control precision.

4. **Evidence correspondence check:** compare the specification with the supplied rule record.
5. **Sentence correspondence check:** parse the generated CNL with a fixed grammar for the specification, reconstruct the supported fields, and compare them with the original specification.

Evidence correspondence, sentence correspondence, and boundary probes for logic–rule comparison were added for the fault-injection evaluation below; they were not applied in the reported training runs.

Fault-injection evaluation. To assess which check detects which error, we inject faults into the pedestrian-entry and position-hold specifications used to produce learning inputs. Each task contributes 24 base cases: 12 archived evaluation contexts paired with two supported durations (0.5/1.5 s for entry and 0.5/2.0 s for holding). These are repeated contexts under two rule settings, not 24 distinct specification parameterizations or independent road scenes. Before injection, we verify the archived specifications and P2 text against the maintained constructors and renderers. The evaluation reuses the unaltered structural validators, logical lowering, renderers, and SMT compiler.

Each base case yields one valid case, six supported faults, and one control in which the source record and the specification are changed to the same wrong value: 192 cases per task and 384 in total. The faults change a supported duration in the specification only, reverse a comparison, omit a release condition, change a unit without rescaling, or alter the sentence's duration or negation. For entry, release omission removes the entry gate. For holding, it keeps the distance restriction active after the hold ends, and comparison reversal affects the inclusive hold endpoint. The probes comprise 7,392 concrete queries; UNKNOWN outcomes are recorded separately. Detection results are reported in Section 5.9 (Table 15).

Sentence correspondence is limited to each specification's fixed grammar. Fixed patterns constrain the target, units, negation, and relation; the parser extracts the duration and reconstructs the supported fields. This is not a general test of semantic equivalence for arbitrary language. An earlier fault-injection evaluation on a separate pedestrian-entry specification, developed for the expression-diversity study, uses probes at $-100, 0, +100$ ms (192 cases) and is described in Supplementary Section S4.

Scope of the checks. Check conclusions hold only within supported specification structures, supplied inputs, explicit assumptions, and the finite logical scope. The accident examples in the Introduction motivate checking interacting requirements, but the reported experiments assess the supported task specifications rather than reconstructing those accidents or verifying a general composition of traffic rules and vehicle dynamics. Consistency of a specification alone does not establish safe behavior or progress toward an objective. Pedestrian entry assumes a single clearance transition without reoccupation; position holding checks only listed coordinate samples. Evidence records and internal consistency do not establish observation truth, rule suitability, or road safety, and logic–rule comparison establishes correspondence between two implementations of the same supplied rule.

Specification checking and model evaluation measure different objects. The former assesses correspondence among structure, evidence, logic, and language; the latter measures whether the selected candidate is admissible and preferred. A model can select a prohibited candidate even when its specification passes the checks, so Section 5 reports the two with their respective evaluation units and denominators.

3.6 Composition of Supplied Constraints

The composition diagnostic checks whether separately supplied conditions can hold together under an explicit shared context. Let Φ_E encode supplied observations and modeling assumptions, and let Φ_j encode source constraint j. For a selected source set Γ, the query is

$$\Phi_E \wedge \bigwedge_{j \in \Gamma} \Phi_j. \tag{8}$$

SAT provides an assignment satisfying the selected conditions together; UNSAT identifies an incompatible combination relative to Φ_E. The context is checked first. Every source constraint is then checked separately, in pairs, and in the full combination. Enumerating subsets for these two- or three-source cases identifies subset-minimal conflict groups: the group is UNSAT and every proper subset is SAT. This identifies incompatible requirements, not which source should be changed.

Documented conditions are kept separate from authored states, candidate requests, modeling assumptions, and test objectives. We manually reviewed official accident investigations, operating guides, traffic guidance, and regulations; the selected cases use NTSB, FHWA, Waymo, and UK Department for Transport documents. The constraints are manually formalized, then executed by the unchanged platform Core/SMT compiler. Supplementary Section S13 lists every natural-language statement with its source role and formal condition; these statements explain the formalization and are not outputs of an unrestricted parser or the CNL correspondence evaluation.

The braking case selects the historical suppression branch with an already recognized emergency, persistent hazard, no takeover, and no alternative path [28]. The NTSB policy record gives a braking delay $t_b \geq 1$ s; an authored constant-deceleration model gives stopping distance $x \geq vt_b + v^2/(2a)$; and an authored test objective requires $x \leq d$. Emergency-trigger eligibility and jerk are outside the model, and the authored terms determine whether the selected delay is compatible with a hypothetical state. The signal case uses FHWA's nominal stopping model and a two-action stop/proceed abstraction [12, 13]. The interlock case checks candidate autonomous driving against the manufacturer's listed blockers [42]; it makes no claim about rolling or towing. The Rule 129 case fixes other exceptions absent and compares a restricted faithful encoding with a deliberately omitted stationary-vehicle exception [10].

Candidate requests are explicit: permission to cross does not require crossing, and a blocked driving candidate does not make holding inconsistent. These distinctions prevent manufactured conflicts from turning optional progress into an unconditional obligation. The diagnostic checks compatibility within the stated abstractions, rather than reconstructing accidents or establishing the safety of an operating system or a complete operating policy. Supplementary Section S13 retains the earlier synthetic boundary diagnostic separately.

3.7 Language Generation and CoC Augmentation

CNL is generated only from structurally validated specifications. Fixed sentence rules express the target, restricted action, condition, value, and unit. Neither reference labels nor progress scores are inputs to the renderer. The pedestrian-entry specification with a 0.5 s wait generates:

> Do not enter the pedestrian conflict zone while it is occupied. Enter only after the zone has remained clear for at least 0.5 s.

The first sentence expresses the occupancy prohibition; the second expresses the release and waiting condition (Table 3). "At least" includes the boundary, and "only after" states a necessary condition for entry rather than an instruction to enter at a particular time. Generated sentences follow the original CoC, before candidate descriptions and the selection instruction. Fig. 2 reproduces two complete model inputs. Clearance time is supplied separately, so this text does not give the answer time directly. The EBLC specification, check verdicts, and evidence records are not inserted into the model prompt.

Comparison conditions. P0 uses the original CoC, P1 adds a directly written natural-language guard, and P2 adds CNL generated from an EBLC specification. For pedestrian entry, P1 uses a template from the preceding study that produces language directly from rule values [21]:

> Do not enter while the pedestrian conflict zone is occupied. After it becomes clear, remain stopped for at least 0.5 s before entering.

The temporal candidates specify entry times or non-entry, so P1 and P2 induce the same admissible set in this task. However, "remain stopped" is stronger: it also restricts motion without entry. We do not claim equivalence for general trajectories.

For position holding, the original CoC is:

> Remain at the current position until the required hold period has ended, then proceed when permitted.

Its P1 natural-language guard uses a template written for this task:

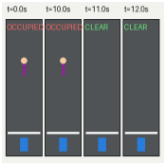
> For the first 2.0 seconds, keep the vehicle within 0.200 metres of its starting point at all listed times, including the final time of this waiting period. Moving farther away is allowed only after this period ends.

Its P2 sentences are generated from the position-hold specification (Table 3; Fig. 2b). P1 and P2 share a learning interface; their comparison examines differences in wording and construction path. Rule availability in each study is summarized in Table 6.

Scope of sentence correspondence. The logical conditions for static and maneuver tasks do not classify non-completing candidates as violations, but their generated sentences do not express this exemption. Sentence–logic correspondence in these tasks is therefore limited to numerical bounds for completing candidates.

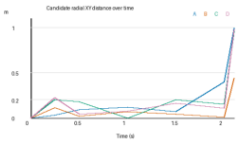
Use of AI-assisted tools. OpenAI Codex was used to assist with English translation and revision of the manuscript and with implementation and checking of supporting scripts. The authors retain responsibility for the scientific content, analyses, and final manuscript. This assistance is separate from the Qwen and Alpamayo models evaluated in the experiments.

(a) Pedestrian clearance, 0.5 s rule




```
Requirement CoC: Yield to the crossing pedestrian, then proceed
through the intersection.
Do not enter the pedestrian conflict zone while it is occupied. Enter
only after the zone has remained clear for at least 0.5 s.
Candidate A: enters the intersection at t=10.5 s and completes the
route; route progress 110.0 m.
Candidate B: enters the intersection at t=11.5 s and completes the
route; route progress 100.0 m.
Candidate C: enters the intersection at t=12.5 s and completes the
route; route progress 85.0 m.
Candidate D: remains stopped through the horizon; route progress 0 m.
The storyboard is ordered by time and shows when the conflict zone
becomes clear. Select the admissible candidate with greatest route
progress. Answer exactly A, B, C, or D.
Reference label: B
```

(b) Position hold, 2.0 s rule, interval-maximum representation



```
Requirement CoC: Remain at the current position until the required
hold period has ended, then proceed when permitted.
Constraint:
Hold the ego vehicle's initial position for at least 2.0 s. At every
listed waypoint with t <= 2.0 s, including the endpoint, the radial XY
distance from the initial position must be at most 0.200 m. Movement
after that period is permitted.
All candidates start at (x,y)=(0,0) m at t=0 s. The diagram shows
candidate radial distances over time. Numerical values below define
the candidates.
Candidate A: maximum radial XY distance for 0 <= t <= 0.5 s: 0.092801
m; for 0 <= t <= 2.0 s: 0.398600 m. Progress score: 45.
Candidate B: maximum radial XY distance for 0 <= t <= 0.5 s: 0.113600
m; for 0 <= t <= 2.0 s: 0.113600 m. Progress score: 24.
Candidate C: maximum radial XY distance for 0 <= t <= 0.5 s: 0.199000
m; for 0 <= t <= 2.0 s: 0.200000 m. Progress score: 25.
Candidate D: maximum radial XY distance for 0 <= t <= 0.5 s: 0.222000
m; for 0 <= t <= 2.0 s: 0.222000 m. Progress score: 111.
Select the candidate that satisfies the constraint and has the
greatest progress score among permitted candidates. Progress is a
supplied synthetic utility. Answer exactly A, B, C, or D.
Reference label: C
```

Fig. 2 Two archived test inputs. (a) Pedestrian entry with a 0.5 s rule. (b) Position hold with a 2.0 s rule and interval-maximum candidate descriptions. Bold green text is the CNL generated from an EBLC specification (P2); P0 inputs omit it. Text is verbatim except for line breaks added around the CNL. The model outputs one candidate label; the reference label is shown for illustration and is not part of the input.

4 Experimental Setup

This section defines two evaluation tracks: constraint compliance and preferred candidate selection by the model, and checking the specifications used to construct learning inputs. It defines the tasks and reference labels, the metrics and aggregation units, and the training and comparison protocols. All learning experiments use synthetic candidate selection; the protocol of the preliminary Alpamayo-R1 diagnostics is given in Supplementary Section S1.

4.1 Tasks and Data

Each example contains an image, the original CoC, a constraint sentence when provided by the condition, descriptions of the supplied candidates, and a selection instruction (Fig. 2). Candidate measurements are supplied in text; the VLM does not

Table 5 Synthetic dataset sizes per input condition. A dash denotes no separate validation split.

Task family	Training	Validation	Primary test
Pedestrian entry	960	240	240
Static (three tasks)	576	–	144
Maneuver (two tasks)	384	–	96
Position hold	960	128	256

estimate them from the image. Static tasks have two candidates; pedestrian entry, maneuver tasks, and position holding have four.

Six behavioral tasks. The six tasks form three families. Static tasks apply maximum-speed, minimum-clearance, and minimum-entry-delay rules to numerical candidate summaries. Pedestrian-entry examples vary the clearance time, required wait, candidate entry times, and label assignments. Maneuver tasks concern stopping and lane changing. Examples occur in pairs with the same scene and candidates but different supplied rules, so the preferred candidate can change: entry 0.5 s after a zone clears is admissible under a 0.5 s waiting rule but prohibited under a 1.5 s rule.

Position holding. This task uses the position-hold constraint in Section 3.4. Synthetic candidate profiles are presented as XY coordinates or as interval-maximum distances calculated from the same coordinates; images are diagrams of the profiles. Each profile yields eight rows: two label rotations, two hold durations, and two representations. The training, validation, and test splits contain 120, 16, and 32 profiles, with no profile shared across splits.

Table 5 gives rows per input condition. P0, P1, and P2 share examples and splits, differing only in their constraint text. Rows are constructed examples rather than independent road videos. Pedestrian-entry images do not overlap across splits, whereas all 13 static primary-test images also occur in training and maneuver data reuse two diagrams. Static and maneuver results therefore measure the use of supplied constraints rather than visual generalization. Split inventories are given in Supplementary Section S2.

Preferred candidates and reference labels. As defined in Section 3.1, the preferred candidate has the highest supplied progress score among admissible candidates; its label is the reference label. Admissibility is computed from the supplied rule and candidate values by integer calculations implemented separately from the logical encoding; position holding checks $x_i^2 + y_i^2 \leq 200^2$ (mm) for every sample up to the hold endpoint. SMT candidate verdicts are compared with these calculations (Section 5.9) and kept as metadata rather than placed in the prompt. The supervised response contains only the reference label. Progress scores are supplied synthetic values whose ordering among admissible candidates determines the preferred candidate. Labels rotate within each task's label set: A–B for static tasks and A–D for the other tasks (Supplementary Section S2). Score variation and label rotation reduce fixed associations between the answer and a label, position, or score, but do not remove shortcuts based on the supplied numerical values.

4.2 Metrics and Aggregation

Preferred-selection score (PSS) assigns one to a prediction that selects the reference label of the highest-scoring admissible candidate, and zero to every other output. The mean is reported as a percentage. It measures how often the preferred candidate is selected, not the size of the progress-score difference when another candidate is selected. **Constraint violation rate (CVR)** is the percentage of all predictions that select a prohibited candidate. An admissible candidate with a lower progress score receives zero for preferred selection but is not a violation. The two metrics therefore need not sum to 100%. **Rule-pair accuracy (RPA)** is the percentage of same-scene, same-candidate pairs with different supplied rules for which both reference labels are selected. Some position-hold pairs share a reference label, so this metric does not require a changed choice in every pair. **Constraint-compliant completion** is the percentage completing the task without selecting a prohibited candidate; it is auxiliary. Unextractable answers receive zero preferred-selection credit and count as neither violations nor admissible choices; their rate is reported separately. Thus, the complement of the constraint violation rate is not the admissible-choice rate when extraction failures occur. Position-hold violations use the strict 0.2 m radius, with a 0.01 m tolerance reported only as an auxiliary measure.

Reported seed standard deviations use the sample convention $(n-1)$. Fresh-adapter results report means and sample standard deviations over three seeds. Each condition combines both durations, yielding 720 pedestrian-entry predictions (240 rows $\times$ three seeds) or 384 position-hold predictions per representation (128 rows $\times$ three seeds). Continuation results report counts per duration and representation (192 predictions from 64 rows and three seeds). Six-task and additional evaluations report three-seed means. Supplementary Section S6 gives individual seeds for the fresh-adapter, six-task, representation, and robustness comparisons; reasoning-prompt and exploratory diagnostics are detailed in Supplementary Sections S8 and S10. Seeds repeat optimization and do not add independent examples.

Specification checks comprise the fault-injection evaluation of Section 3.5 (384 cases, 7,392 queries), the 5,376 position-hold candidate checks, and, for the pedestrian-entry specification, one satisfiability query and 24 boundary-candidate queries for each of four waits (0.5, 1.5, 0.8, and 1.8 s), 100 queries in total. These units are reported separately from each other and from model-prediction denominators.

4.3 Model and Training Settings

We fine-tune Qwen3-VL-2B-Instruct [32], a model small enough for every comparison to be repeated with fresh adapters over three seeds under matched budgets; a larger model of the same family is evaluated without training (Section 4.4). Adaptation uses LoRA [19] (rank 8, $\alpha = 16$, dropout 0.05) with AdamW (learning rate 2×10^{-4}, weight decay 0.01), batch size 2, and gradient accumulation over eight steps. Each fit runs for three epochs with optimization seed 42, 17, or 123; pedestrian-entry and position-hold fits perform 180 updates, and static and maneuver fits 108 and 72. Within each task family, compared conditions share examples, epochs, optimizer settings, and the final-checkpoint policy; validation scores do not select checkpoints.

Table 6 Training and evaluation information in the three studies. Fresh adapters share their initialization within each task and seed.

Study	Training	Evaluation	Question
Fresh-adapter study	Same pretrained base; CoC alone or CoC + CNL, each with or without semantic loss	(a) Training input type; (b) same CoC + CNL for all adapters	With evaluation information and initialization held fixed, do training constraints reduce violations and improve rule-dependent selection?
Continuation study	Existing pedestrian-entry adapters; P0: CoC, P1: CoC + natural-language guard, P2: CoC + CNL	Same CoC + P2 CNL for every adapter, before and after continuation	Does a constraint-bearing training history improve behavior under identical evaluation inputs?
Six tasks	P0: CoC; P1 and P2: CoC + respective constraints	Same input type as training	Do models use supplied constraints and change their choices when the rule changes?

Input lengths differ across text conditions. Fine-tuned adapters use greedy decoding with a 12-token answer limit.

4.4 Comparison Designs

Table 6 distinguishes three studies by the information available during training and evaluation. Models trained with constraint text also receive it at evaluation, matching the intended use of case-specific constraints (Section 6.3).

Fresh-adapter study. We train new LoRA adapters from the same pretrained base on pedestrian entry and position holding. The 2×2 design crosses CoC-only versus CoC + CNL inputs with cross-entropy (CE) alone versus CE plus semantic loss; seed-specific initial LoRA tensors are identical across the four conditions. Two tasks, four conditions, and three seeds yield 24 fits. Our semantic loss is a categorical adaptation of Xu et al. [45]. At the first supervised answer-token position, let z_a be the logit for label $a \in \mathcal{Y} = \{A, B, C, D\}$:

$$q_a = \frac{\exp z_a}{\sum_{b \in \mathcal{Y}} \exp z_b}, \qquad \mathcal{L}_{\text{train}} = \mathcal{L}_{\text{CE}} - \lambda \log \sum_{a \in \mathcal{Y}_C} q_a, \tag{9}$$

where $\mathcal{Y}_C \subseteq \mathcal{Y}$ is the set of labels whose candidates are admissible. We fix $\lambda = 1$ before execution. Even with CoC-only inputs, the loss uses the example's admissible set, so rule information enters through the objective; inference neither filters candidates nor uses SMT. The protocol specifies evaluation with the training input type as primary and common-CNL evaluation as additional; we report both and emphasize the latter when comparing training conditions.

Continuation study. We continue the nine pedestrian-entry adapters (P0, P1, P2 $\times$ three seeds) on 960 position-hold rows for 180 further updates. P0 receives the original CoC at both stages, while P1 and P2 receive their respective constraints at both stages. Before and after continuation, all adapters receive identical inputs with the P2 constraint sentence. The comparison therefore includes the effects of different

Table 7 Comparison conditions and principles shared with related approaches. Correspondences identify selected principles, not reproductions of complete systems.

Condition	Shared principle	Budget
CNL at evaluation for CoC-only-trained adapters	Use of rules at inference; regulation-augmented reasoning also uses rules at this stage [5]	Matched fresh-adapter training; continuation reported separately
CoC + CNL for pretrained 8B	Prompting a larger model with rules	No task-specific training
Categorical semantic loss	Constraints in the training objective [45]	Matched fresh-adapter training
General reasoning prompts	Zero-shot CoT and Plan-and-Solve [22, 40]	No additional training; unequal generation limits
Logical-text control	Formal representation of requirements [7, 8]	Matched pedestrian-entry training
P1 and alternative natural-language guards	Natural-language constraints in training [21]	Matched within each task family

training histories. Previously inspected profiles with prohibited candidates only 1 mm beyond the radius are reevaluated separately as a regression check (Supplementary Section S11).

Identifiability without rule text. When the distinguishing rule is omitted, identical inputs can have different reference labels. Every six-task rule pair has different reference labels, giving a 50% preferred-selection score ceiling for rule-free inputs; in the position-hold set, half the pairs do, giving an empirical ceiling of 75%. Fewer violations without rule text therefore do not show that a model changes its choice with the rule, and the six-task study measures the use of constraints supplied at both training and evaluation rather than a training benefit under identical information. Rule-pair accuracy and the common-CNL evaluation are used for this reason to assess rule-dependent selection directly.

Pretrained-VLM evaluation. Qwen3-VL-2B-Instruct and Qwen3-VL-8B-Instruct [32, 33] receive CoC and CNL without fine-tuning or demonstrations on all 736 primary-test inputs, with one greedy response per input. The primary evaluation fixes a 12-token output limit before execution; after observing truncated explanations, we add a post-hoc evaluation with a 512-token limit and final-answer extraction, and report both, together with output-limit hits and answer-extraction failures. This comparison includes adaptation to the task format and is not a matched-budget training comparison.

4.5 Comparison Conditions and Related Approaches

The systems reviewed in Section 2 use different interfaces: DriveLM and DriveVLM address driving question answering and planning, while nl2spec and NL2TL output temporal-logic formulas rather than action labels [7, 8, 36, 37]. Their native execution paths are documented in Supplementary Section S5; adding our constraints or an output adapter would evaluate modified methods. We instead use controls that share selected principles with prior work (Table 7); they are not reproductions of the cited systems.

4.6 Additional Evaluations

The following evaluations use archived checkpoints, including those from additional conditions trained with matched task-family budgets. Their rows are not added to the primary-test denominator. Procedures are detailed in Supplementary Sections S2, S8, and S10.

- **General reasoning prompts.** Zero-shot CoT (128 reasoning tokens) and basic Plan-and-Solve (256 tokens) [22, 40] are applied to the final P0 checkpoints without rule information or further training. A first stage generates reasoning for each unique input (546 per method), and a second stage extracts the label; each method yields 1,440 predictions over three seeds.
- **Paraphrasing.** A held-out constraint wording is given to P1 and P2 on the 240, 144, and 96 primary-test rows of the three families; P0 inputs, which contain no constraint text, are unchanged. A smaller exploratory diagnostic tests a second wording with 24 rows per wording and seed.
- **Unseen values.** Values and durations absent from training are used, such as pedestrian-entry waits of 0.8 and 1.8 s instead of 0.5 and 1.5 s.
- **Combined difficulty.** A 96-row maneuver set combines negation, clause order, equality boundaries, and m/cm units.
- **Alternative pedestrian-entry representations.** We report two additional conditions trained with the same budget as P0–P2: an alternative natural-language guard and a logical formula supplied as text. Each was evaluated on the 240 primary-test and 240 unseen-wait rows. The logical text is a model input, not a solver or translator output.
- **Image interventions.** Pedestrian-entry adapters receive original, blank, or redrawn images with clearance 1.0 s later, with all other inputs and the original references fixed (24 rows per condition, 648 predictions). Scores on redrawn images measure an input intervention rather than scene-understanding performance; we also report the fraction of choices that change relative to the original image.

4.7 Bounded Constraint-Composition Diagnostic

This diagnostic, run after and independently of the training runs, evaluates the manually formalized source groups in Section 3.6. A source catalog records the document version, URL, paragraph or page, supported interpretation, and scope. Eight official documents were inspected; two further UNECE candidates were excluded because their full official text could not be retrieved. The executed cases use four document families, with scenario states, candidate requests, motion assumptions, and test objectives marked as authored inputs. No operational log is used.

Parameter grids bracket each feasibility threshold by 1 mm: braking (48 cases; speeds of 10/15/20 m/s, decelerations of 5/7 m/s^2, and delays of 0/1000 ms, with zero delay as a control), yellow-interval feasibility (64 cases; nominal deceleration of 3 m/s^2, not an emergency-braking bound), all 16 interlock states with drive/hold candidates (32 cases), and all eight exception states with cross/hold candidates and faithful/omitted-exception encodings (32 cases). Supplementary Section S13 gives the full grids.

The 176 configurations yield 896 queries, covering the context alone and every nonempty source subset. Verdicts are compared with independent rational feasibility calculations, and each SAT witness is checked numerically. For each UNSAT combination, all proper subsets identify subset-minimal conflict groups. We distinguish incompatibility introduced by suppression from avoidance already infeasible without it, and permission to cross from an obligation to cross. These configurations and queries are counted separately from fault injection, VLM predictions, and the earlier 123-case synthetic boundary diagnostic. Supplementary Section S13 records both protocols and their distinct scopes.

5 Results

The behavioral track first tests training with constraints under common-CNL evaluation and training-matched evaluation (Sections 5.1 and 5.2), then examines continuation training (Section 5.3). The six-task and rule-pair results assess use of the supplied rule; reasoning, wording, value, and image controls examine alternative explanations and robustness. The specification track separately reports fault detection and candidate-verdict correspondence (Section 5.9). A separate diagnostic tests composition consistency (Section 5.10). Section 5.11 brings the two tracks together.

Answer-extraction failures are zero in the fresh-adapter, continuation, regression, six-task, robustness, reasoning-prompt, and exploratory image/wording evaluations. The pretrained-model evaluations have nonzero failure rates, reported below and in Supplementary Section S7.

5.1 Constraint Training with Identical Evaluation Inputs

The fresh-adapter study uses two evaluations: inputs matching each training condition and a common-CNL evaluation for every adapter (Section 4.4). This subsection emphasizes the latter, which fixes evaluation information. Section 5.2 reports the same adapters with training-matched inputs, keeping the effects of input information and loss design distinct.

Same initialization and evaluation inputs. Table 8 compares the four fresh-adapter conditions; individual-seed values are given in Supplementary Section S6. In the CE-only comparison, the conditions differ in whether constraint sentences appear in training. With the same CNL at evaluation, CoC-only-trained adapters violate the rule in 17.92% of pedestrian-entry predictions and 6.51% and 6.25% of position-hold predictions with interval maxima and XY coordinates, respectively. Training with CNL reduces these rates to 1.11%, 0.00%, and 1.56%. Preferred-selection score increases from 50.14%, 78.39%, and 75.52% to 98.47%, 96.09%, and 93.49%. With initialization and evaluation information controlled, these differences support the benefit of including constraint text during training.

CoC-only-trained preferred-selection score remains close to the rule-free input ceilings of 50% for pedestrian entry and 75% for position holding, even after CNL is provided at evaluation. These ceilings apply when rule text is absent (Section 4.4); they are not mathematical upper bounds for the common-CNL inputs. Rule-pair accuracy also separates the CE-only conditions: CoC-only-trained adapters reach

Table 8 Fresh adapters with identical CNL at evaluation. Cells show preferred-selection score (first line) and constraint violation rate (second line), in percent, as mean ± standard deviation over three seeds. Column labels describe training inputs.

Task	CoC CE	CoC CE + semantic	CoC + CNL CE	CoC + CNL CE + semantic
Pedestrian entry	50.14 ± 1.46 17.92 ± 8.91	50.00 ± 0.00 0.00 ± 0.00	98.47 ± 1.20 1.11 ± 1.58	97.64 ± 2.14 0.83 ± 1.44
Position hold: maxima	78.39 ± 1.97 6.51 ± 3.52	75.52 ± 0.90 0.26 ± 0.45	96.09 ± 1.35 0.00 ± 0.00	96.09 ± 2.34 0.00 ± 0.00
Position hold: XY	75.52 ± 0.45 6.25 ± 2.71	74.74 ± 0.45 0.52 ± 0.90	93.49 ± 1.19 1.56 ± 0.78	92.71 ± 3.61 2.08 ± 2.96

1.94%, 56.77%, and 51.56%, whereas CNL-trained adapters reach 96.94%, 92.71%, and 86.98%. In these checkpoints, training with constraint text supports stronger rule-dependent selection than inference-time rule provision alone.

VLMs without task-specific training. Table 9 reports Qwen3-VL-2B-Instruct and Qwen3-VL-8B-Instruct with the same CoC and CNL inputs but no task-specific training. Panel (a) reports selection scores together with answer-extraction failures. At the prespecified 12-token limit, every pretrained 2B response reaches the cap and yields no extractable label. Panel (b) reports the post-hoc 512-token evaluation on the three fresh-adapter tasks; extraction failures remain frequent. Full results and cap-hit counts are reported in Supplementary Section S7. The 8B model's primary constraint violation rate is 68.75% for pedestrian entry, 25.00–50.00% for the static tasks and stopping, and 7.03% and 33.59% for the two position-hold representations. Lane changing has no observed violations but 75.00% preferred-selection score.

Only XY-coordinate responses hit the 8B 12-token limit (26/128). With 512 tokens, pedestrian-entry preferred-selection score / violation rate is 27.50% / 66.67%, and XY position holding is 36.72% / 34.38%, with 17.19% extraction failures. Other 8B task scores are unchanged.

The comparison identifies limitations of inference-only rule provision on these inputs; it does not isolate a model-size effect. Because the pretrained 2B evaluation has substantial extraction failures even at 512 tokens, these data also cannot isolate whether CoC-only fine-tuning reduced any prior rule-following ability.

5.2 Constraints in the Loss and in the Input

Table 10 evaluates the four fresh-adapter conditions with inputs matching training. Without rule text, semantic loss reduces pedestrian-entry violations from 19.58% to 0.00%, and position-hold violations from 10.94% to 1.56% with maxima and 0.52% with XY coordinates. However, preferred-selection score stays near the rule-free ceilings of 50% and 75% (Section 4.4), and rule-pair accuracy remains 0% or approximately 50%.

Half the position-hold pairs share a reference label, permitting up to 75% preferred-selection score without distinguishing their rules. Reduced violations therefore do not by themselves establish rule-dependent choice. The results are consistent with more

Table 9 Pretrained 2B and 8B and CNL-trained 2B with CoC + CNL inputs. (a) Preferred-selection score / constraint violation rate / answer-extraction failure rate (%). All use the prespecified 12-token limit; pretrained models give one greedy response per input, and trained scores are three-seed means. (b) Post-hoc 512-token results for pretrained 2B on the fresh-adapter tasks.

(a) Prespecified 12-token evaluation

Task	2B, no task training	8B, no task training	2B, CNL training
Maximum speed	0.00 / 0.00 / 100.00	60.42 / 39.58 / 0.00	100.00 / 0.00 / 0.00
Minimum clearance	0.00 / 0.00 / 100.00	54.17 / 45.83 / 0.00	100.00 / 0.00 / 0.00
Minimum entry delay	0.00 / 0.00 / 100.00	50.00 / 50.00 / 0.00	100.00 / 0.00 / 0.00
Pedestrian entry	0.00 / 0.00 / 100.00	27.08 / 68.75 / 0.00	98.47 / 1.11 / 0.00
Stopping	0.00 / 0.00 / 100.00	50.00 / 25.00 / 0.00	100.00 / 0.00 / 0.00
Lane change	0.00 / 0.00 / 100.00	75.00 / 0.00 / 0.00	97.92 / 0.00 / 0.00
Position hold: maxima	0.00 / 0.00 / 100.00	70.31 / 7.03 / 0.00	96.09 / 0.00 / 0.00
Position hold: XY	0.00 / 0.00 / 100.00	35.94 / 33.59 / 20.31	93.49 / 1.56 / 0.00

(b) Pretrained 2B, post-hoc 512-token evaluation (%)

Task	Preferred-selection score	Violation rate	Extraction failures
Pedestrian entry	10.83	11.25	65.42
Position hold: maxima	14.84	0.00	79.69
Position hold: XY	7.03	7.03	75.78

Unextractable answers receive zero preferred-selection credit and count as neither violations nor admissible choices; the pretrained 2B model's zero 12-token violation rate therefore does not indicate compliance. Admissible lower-scoring choices also receive zero preferred-selection credit, so the three rates need not sum to 100%. Trained pedestrian-entry and position-hold rows use fresh CE adapters; other trained rows use six-task P2 checkpoints. This is not a matched-training comparison across model sizes. Panel (b) changes the output limit and extraction protocol (Supplementary Section S7).

conservative selections that are admissible under either rule, rather than improved discrimination between rules. Providing CNL only at evaluation does not close the preferred-selection score gap for the semantic-only-trained condition (Table 8).

With CNL training inputs, adding semantic loss gives no consistent further benefit: preferred-selection score changes by −0.83 to 0.00 percentage points and violation rate by −0.28 to +0.52 points. These three-seed estimates do not establish equivalence between the losses. In this setting, the loss penalty mainly reduces violations without supplying the rule-dependent input distinction, whereas training with rule text supports that distinction.

5.3 Position-Hold Continuation

The continuation study gives a similar direction of benefit under identical evaluation inputs (Table 11; Section 4.4). Before position-hold training, adapters with constraint-bearing histories already make fewer violations when candidates are represented by interval maxima. For the 2.0 s rule with interval maxima, P0, P1, and P2 violate the rule in 64, 23, and 20 of 192 predictions. After continuation, the counts are 22 (11.46%), 1 (0.52%), and 0. P2 preferred-selection score rises from 81.77% to 98.44%, and P1 reaches the same preferred-selection score with one violation. Every condition receives the same rule at evaluation, so the differences cannot be attributed to

Table 10 Fresh adapters evaluated with their training input type. Cells show preferred-selection score and constraint violation rate (%, mean ± seed standard deviation) on the first two lines; parentheses give rule-pair accuracy (%). CE denotes cross-entropy loss; semantic denotes the additional constraint loss defined in Eq. (9).

Task	CoC CE	CoC CE + semantic	CoC + CNL CE	CoC + CNL CE + semantic
Pedestrian entry	49.86 ± 0.24 19.58 ± 6.14 (0.00)	50.00 ± 0.00 0.00 ± 0.00 (0.00)	98.47 ± 1.20 1.11 ± 1.58 (96.94)	97.64 ± 2.14 0.83 ± 1.44 (95.28)
Position hold: maxima	75.00 ± 0.00 10.94 ± 1.35 (50.00)	74.48 ± 0.90 1.56 ± 1.56 (49.48)	96.09 ± 1.35 0.00 ± 0.00 (92.71)	96.09 ± 2.34 0.00 ± 0.00 (92.71)
Position hold: XY	72.66 ± 1.35 10.94 ± 2.71 (48.44)	74.74 ± 0.45 0.52 ± 0.90 (50.00)	93.49 ± 1.19 1.56 ± 0.78 (86.98)	92.71 ± 3.61 2.08 ± 2.96 (85.42)

Table 11 Position-hold continuation: strict violation counts and preferred-selection score (%) before → after training. Each cell summarizes 192 predictions (64 rows over three seeds). Evaluation inputs are identical across conditions.

	Violations			Preferred-selection score (%)		
Representation, hold	P0	P1	P2	P0	P1	P2
Maxima, 0.5 s	29 → 0	13 → 1	7 → 1	47.92 → 67.19	56.25 → 89.58	52.08 → 94.79
Maxima, 2.0 s	64 → 22	23 → 1	20 → 0	63.02 → 86.98	85.94 → 98.44	81.77 → 98.44
XY, 0.5 s	56 → 1	49 → 5	53 → 3	47.92 → 61.98	51.56 → 81.25	50.00 → 90.63
XY, 2.0 s	121 → 23	126 → 16	115 → 4	32.81 → 88.02	33.85 → 91.15	39.58 → 97.92

withholding it from P0 at that stage. They reflect the compared training histories, including whether constraint sentences were supplied at both training stages.

With XY coordinates and the 2.0 s rule, P2 preferred-selection score rises from 39.58% to 97.92%, but four violations remain. Preferred-selection score and constraint compliance are distinct: under the 0.5 s rule with interval maxima, P0 makes no violations yet reaches only 67.19% preferred-selection score, often selecting admissible candidates with lower progress.

On the new position-hold test set, a 0.01 m tolerance leaves violation counts unchanged for both the fresh-adapter and continuation studies: every prohibited candidate exceeds the radius by at least 22 mm. The tolerance does affect the separate regression diagnostic, as reported in Supplementary Section S11.

The earlier regression diagnostic remains difficult after training. Its XY/2.0 s stratum still yields 106, 79, and 77 violations out of 192 for P0, P1, and P2. The smallest prohibited overshoot is 1 mm in that diagnostic, compared with 22 mm in the new primary test. The sets are not pooled; strict and tolerance-based regression results are reported separately in Supplementary Section S11. The new test supports improved synthetic selection, with remaining sensitivity to representation and small boundary overshoots.

Table 12 Six-task preferred-selection score / constraint violation rate (%), three-seed means. These checkpoints are distinct from the fresh-adapter study in Tables 8 and 10.

Task	P0	P1	P2
Maximum speed	50.00 / 24.31	100.00 / 0.00	100.00 / 0.00
Minimum clearance	50.00 / 24.31	100.00 / 0.00	100.00 / 0.00
Minimum entry delay	50.00 / 24.31	100.00 / 0.00	100.00 / 0.00
Pedestrian entry	49.86 / 19.17	96.81 / 1.39	98.06 / 0.83
Stopping	50.00 / 23.61	100.00 / 0.00	100.00 / 0.00
Lane change	50.00 / 38.19	100.00 / 0.00	97.92 / 0.00

Table 13 Six-task rule-pair accuracy (%), three-seed means. Static and maneuver values aggregate their subtasks.

Family	P0	P1	P2
Pedestrian entry	0.00	93.61	96.11
Static	0.00	100.00	100.00
Maneuver	0.00	100.00	97.92

5.4 Use of Supplied Constraints Across Six Tasks

Table 12 reports the six-task comparison. P0 lacks rule information and remains near its design-imposed 50% ceiling. Both P1 and P2 have higher preferred-selection scores and lower constraint violation rates in every task. Their preferred-selection scores range from 96.81% to 100%, with violations at or below 1.39%. Pedestrian-entry preferred-selection score exceeds P0 by 46.94 and 48.19 percentage points for P1 and P2, while violations decrease by 17.78 and 18.33 points. P1 and P2 tie on the three static tasks and stopping.

P2 lane-change errors select admissible but lower-progress candidates; constraint-compliant completion is 100%. P2 pedestrian-entry constraint-compliant completion is 99.17%. Supplying constraints in both training and evaluation inputs raises every task to near-ceiling preferred-selection score. Individual-seed values are given in Supplementary Section S6.

5.5 Selection under Rule Changes

Changing the rule while holding scene and candidates fixed tests whether choices change accordingly. Table 13 reports rule-pair accuracy: P0 reaches 0% in all three families, whereas P1 and P2 reach 93.61–100% and 96.11–100%. Constraint-conditioned models change their choices with the rule despite unchanged visual input; P0's result reflects its missing rule information.

In one recorded pedestrian-entry case, clearance occurs at 11.0 s and a 1.5 s wait permits entry from 12.5 s; P0 selects entry at 11.5 s, whereas P1 and P2 select the boundary entry at 12.5 s. This and a P2 violation case are described in Supplementary Section S9.

Table 14 Changed wording and values: preferred-selection score / constraint violation rate (%), three-seed means. Paraphrases reuse primary-test cases.

Evaluation	P0	P1	P2
Pedestrian entry: primary	49.86 / 19.17	96.81 / 1.39	98.06 / 0.83
Paraphrase	49.86 / 19.17	90.56 / 5.00	96.94 / 1.25
Unseen wait	49.86 / 18.47	96.67 / 2.22	96.67 / 1.53
Static: primary	50.00 / 24.31	100.00 / 0.00	100.00 / 0.00
Paraphrase	50.00 / 24.31	93.29 / 0.00	80.32 / 7.87
Unseen values	50.00 / 22.92	99.54 / 0.00	97.22 / 0.23
Maneuver: primary	50.00 / 30.90	100.00 / 0.00	98.96 / 0.00
Paraphrase	50.00 / 30.90	100.00 / 0.00	97.92 / 2.08
Unseen values	50.00 / 30.90	100.00 / 0.00	98.96 / 0.00
Combined difficulty	50.00 / 30.21	94.10 / 0.69	91.32 / 1.04

5.6 General Reasoning-Prompt Controls

With the rule omitted, general reasoning prompts leave P0 preferred-selection score and rule-pair accuracy unchanged (Section 4.6). Across the 480 primary-test rows, P0, Zero-shot CoT, and Plan-and-Solve all reach 49.93% preferred-selection score, with violation rates of 23.06%, 22.64%, and 22.71%, and 0% rule-pair accuracy in every family. All 546 unique Zero-shot CoT first-stage outputs and 498 of 546 Plan-and-Solve outputs are short label strings, so the prompts do not establish multistep reasoning in these label-trained checkpoints. Because identical P0 inputs carry conflicting reference labels under different rules, no fixed output can exceed 50% preferred-selection score. Detailed outputs are given in Supplementary Section S8.

5.7 Representation, Wording, and Unseen Values

Table 14 reports changed wording and numerical values. Paraphrase rows retain the primary-test cases and rules, so differences measure wording sensitivity rather than performance on additional independent scenes. Both constraint conditions substantially outperform P0 on unseen values. For pedestrian-entry paraphrases, P2 retains 96.94% preferred-selection score and 1.25% violations, compared with 90.56% and 5.00% for P1.

Static paraphrases reverse the ordering: P2 preferred-selection score falls from 100% to 80.32%, with 7.87% violations, while P1 reaches 93.29% with no violations. Both the favorable pedestrian-entry result and the unfavorable static result are needed to characterize the CNL interface. On the combined maneuver set, P1 reaches 94.10% preferred-selection score and 0.69% violations, compared with 91.32% and 1.04% for P2. P2 rule-pair accuracy falls from 97.92% to 82.64%. The combined test cannot identify whether negation, clause order, equality, or units individually cause the errors.

A smaller exploratory diagnostic (24 rows per wording and seed) shows the same fragility: a second paraphrase lowers P1 and P2 preferred-selection score to 76.39% and 77.78% (Supplementary Section S10).

Alternative expressions of the same pedestrian-entry rule also perform well on the primary test (individual seeds in Supplementary Section S6). An alternative natural-language guard matches P2 preferred-selection score at 98.06% (1.11% violations), and logical text, a model-input control rather than a solver or translator output, reaches 99.17% (0.14%). On unseen waits, the natural-language guard reaches 98.61% (0.56%), whereas logical-text preferred-selection score falls to 81.25% despite only 0.83% violations. Low violation does not imply selecting the highest-progress admissible action. High primary-test scores thus hold across several rule representations, while robustness varies with wording and unseen values.

5.8 Dependence on Visual Input

Image interventions test whether selections change when visual input is removed or conflicts with the retained case (Section 4.6). Although candidate values are supplied in text, choices depend on the image (three-seed means; full results in Supplementary Section S10). P2 preferred-selection score against the original reference falls from 98.61% to 76.39% with a blank image and 79.17% with the redrawn image, and its choice changes on 25.00% and 19.44% of the paired cases. P1 falls from 100% to 81.94% and 87.50%, with choice changes on 18.06% and 12.50%; P0 stays at 50%.

P2 violations remain at 1.39% in every image condition. Preferred-selection score loss therefore does not imply more prohibited choices; changes to admissible but lower-progress candidates also contribute. The 24 rows comprise three base case configurations, each paired with two waiting rules and four label rotations (Supplementary Section S10). With three training seeds, this is an exploratory input-dependence diagnostic.

5.9 Specification Checking

Table 15 reports fault injection into the two specifications used for learning (protocol in Section 3.5). All 48 valid cases pass, and the full suite detects all 288 supported faults: the structural, satisfiability, and logic–rule checks detect 192, and sentence correspondence detects the remaining 96 CNL faults. No fault made a specification unsatisfiable, so the satisfiability check flagged none. An error can trigger several layers, so layer counts are not summed. None of the 7,392 probe queries returned UNKNOWN. All 48 jointly wrong evidence–specification controls remain undetected. If the rule record and its specification field change to the same wrong value, evidence correspondence and the logic–rule comparison still agree and the specification remains satisfiable, so detecting such errors requires evidence independent of the supplied record. An earlier evaluation on a separate pedestrian-entry specification shows the same correspondence between fault types and detecting layers (Supplementary Section S4).

Two further checks use different units from the fault-injection case counts. SMT verdicts agree with the independent integer admissibility calculation on all 5,376 position-hold candidate checks across training, validation, and test (Section 4.1). This measures candidate-level correspondence with the supplied rule.

Table 15 Fault injection into the specifications used for learning. Each row has one case per base context (24 per task). Counts are accepted cases for valid specifications and detected cases otherwise; they are cases, not independent scenes. Detecting-layer names abbreviate the five checks in Section 3.5.

Case	Injection location	Detecting layer	Entry	Hold
Valid specification	None	None (accepted)	24/24	24/24
Changed specification duration	Specification; source unchanged	Logic–rule, evidence	24/24	24/24
Reversed comparison	Logical formula	Logic–rule	24/24	24/24
Deleted release condition	Logical formula	Logic–rule	24/24	24/24
Changed unit	Specification	Structural	24/24	24/24
Changed sentence duration	CNL	Sentence	24/24	24/24
Changed sentence negation	CNL	Sentence	24/24	24/24
Jointly wrong evidence and specification	Source record and specification	Not detected	0/24	0/24
Supported faults: structural, satisfiability, logic–rule			96/144	96/144
Supported faults: full suite			144/144	144/144

For the pedestrian-entry specification, four wait values (0.5, 1.5, 0.8, and 1.8 s) yield 100 queries: four satisfiability and 96 boundary-candidate queries. All match their expected verdicts: 76 SAT, 24 UNSAT, and no UNKNOWN. These are counted separately from the 7,392 concrete queries in the fault-injection evaluation. To illustrate the inclusive boundary, with clearance at 11.000 s and a 0.500 s wait, entry at 11.499 s is UNSAT, whereas 11.500 s and 11.501 s are SAT.

The checks identify supported disagreements among specifications, logical restrictions, sentences, and supplied rules.

5.10 Constraint-Composition Diagnostic

Table 16 reports the document-grounded diagnostic (Sections 3.6 and 4.7). All 176 contexts and individual source groups are satisfiable. Joint checks identify 58 UNSAT and 118 SAT configurations. All 896 subset queries agree with the independent feasibility reference, with zero UNKNOWN outcomes; independent numeric checks validate all 832 SAT witnesses. These counts cover distinct compatibility questions, rather than 58 conflicting real-world policies. The 58 UNSAT configurations comprise 22 requirement conflicts (12 three-way braking and 10 yellow-interval), 6 objectives already infeasible without the delay policy, 22 candidate requests correctly blocked by a source condition, and 8 crossings rejected only by the omitted-exception fault.

The suppression family contains 12 three-way conflicts with every pair SAT. For example, the authored state $v = 10$ m/s, $a = 5$ m/s^2, and $d = 15$ m combines $t_b \geq 1$, $x \geq 10t_b + 10$, and $x \leq 15$. Each pair is satisfiable; together they are inconsistent. Removing the delay admits stopping at 10 m in this simplified model. Six other configurations have an incompatible stopping model and objective even without suppression; their minimal groups correctly exclude the delay policy. This distinction prevents attributing already-infeasible avoidance to a policy interaction. These are

Table 16 Document-grounded conditions combined with authored states, models, objectives, or candidate requests. Counts are diagnostic configurations, not independent road scenes. Individual groups are SAT. Pairwise detection counts UNSAT full configurations having an UNSAT two-group subset. The exception row includes a deliberately faulty transcription and permitted/blocked candidate controls.

Composition	Cases	Full SAT	Full UNSAT	Pairwise detected
Suppression + model + stopping goal	48	30	18	6
Yellow stop/proceed feasibility	64	54	10	10
Autonomous-driving interlock + candidate	32	17	15	15
Crossing exception + candidate	32	17	15	15
Total	176	118	58	46

hypothetical states, not the Tempe collision: NTSB reports that avoidance already exceeded ADS braking specifications at emergency recognition [28]. Yellow-interval conflicts arise when neither the stop nor the proceed branch is feasible under the nominal model (example in Supplementary Section S13).

All 16 interlock hold controls are SAT, while autonomous-driving requests conflict in the 15 states with at least one blocker. With all three Rule 129 exception conditions true, the faithful encoding accepts a crossing candidate and the omitted-exception variant rejects it. Both encodings accept the non-crossing candidate; all 16 such controls are SAT. The exception result identifies an overly restrictive transcription, not a contradiction in the original rule. A single UNSAT verdict for a crossing candidate has the same form whether the candidate is correctly blocked or a permission was lost; comparing the faithful and omitted-exception encodings on the same candidate exposes the lost permission. Thus joint checks, read with explicit source roles and this encoding comparison, distinguish incompatible requirements, blocked candidates, and a lost permission. Supplementary Section S13 reports the earlier 123-case synthetic boundary diagnostic separately.

5.11 Summary of Findings

Table 17 collects the central contrast. Every condition receives the same CoC + CNL inputs at evaluation; only training with constraint text yields both few violations and high rule-pair accuracy among the conditions shown.

- **Training with constraints.** Relative to CoC-only training, CNL training lowers violations from 17.92% to 1.11% on pedestrian entry and raises rule-pair accuracy from 1.94% to 96.94% (Section 5.1). Continuation training shows the same direction: with the 2.0 s rule and interval maxima, P0, P1, and P2 retain 22, 1, and 0 violations of 192 (Section 5.3).
- **Alternatives without training-time rule text.** Semantic loss removes most violations but leaves preferred-selection score near the rule-free input ceilings of 50% and 75% under training-matched evaluation; the pretrained 8B model violates the rule in up to 68.75% of predictions; general reasoning prompts leave P0 at 49.93% (Sections 5.1, 5.2, and 5.6).

Table 17 Central comparison with identical CoC + CNL evaluation inputs: preferred-selection score / constraint violation rate / rule-pair accuracy (%). Fine-tuned rows are three-seed means of fresh 2B adapters (Tables 8 and 10; Supplementary Section S6); pretrained rows use the prespecified 12-token limit (Table 9).

Training	Pedestrian entry	Hold: maxima	Hold: XY
Rules supplied only at evaluation			
None (pretrained 2B)	0.00 / 0.00 / 0.00[a]	0.00 / 0.00 / 0.00[a]	0.00 / 0.00 / 0.00[a]
CoC, CE	50.14 / 17.92 / 1.94	78.39 / 6.51 / 56.77	75.52 / 6.25 / 51.56
CoC, CE + semantic loss	50.00 / 0.00 / 0.00	75.52 / 0.26 / 51.56	74.74 / 0.52 / 50.00
None (pretrained 8B)	27.08 / 68.75 / 0.83	70.31 / 7.03 / 51.56	35.94 / 33.59 / 23.44
Constraint text supplied during training			
CoC + CNL, CE	**98.47 / 1.11 / 96.94**	**96.09 / 0.00 / 92.71**	**93.49 / 1.56 / 86.98**

[a]Answer-extraction failure is 100% at the 12-token limit; the zero violation rate does not indicate compliance (Table 9). 8B rule-pair accuracy is taken from the evaluation record (Supplementary Section S7).

- **Use of supplied constraints.** P1 and P2 reach 96.81–100% preferred-selection score on the six tasks, compared with about 50% for P0, and change their choices when the rule changes (Sections 5.4 and 5.5). P2 is not uniformly better than P1: on static paraphrases, P2 reaches 80.32% versus 93.29% for P1 (Section 5.7).
- **Checkability.** The full check suite detects all 288 supported faults without rejecting the 48 valid cases, but misses all 48 jointly wrong evidence–specification controls (Section 5.9).
- **Composition consistency.** A separate document-grounded diagnostic finds 58 incompatible configurations despite satisfiable individual groups; 12 require checking three sources together, beyond pairwise checks (Section 5.10).

6 Discussion

The discussion follows the three problems in the Introduction: constructing and checking source-bound constraints, preserving their meaning in learning sentences, and using them in action selection. The distinction between constraint compliance and preferences among admissible candidates applies throughout; a high preferred-selection score concerns the supplied candidate set and objective.

6.1 Constraint Construction and Specification Checkability

Reinforcement learning can improve constraint compliance and action selection, but the rules and evaluation criteria used for learning require separate validation: a misspecified rule or reward can yield a policy that optimizes the supplied signal while departing from the intended requirements. In a traffic-merging simulation, Pan et al. report that an agent optimizing a proxy reward based on average vehicle speed learned to stop its vehicle instead of merging, increasing the proxy reward while also increasing mean commute time [31]. When several rules and modeling assumptions apply, their joint consistency and whether their combination permits necessary safety responses

also require examination. This need to check the learning specification applies to both supervised and reinforcement learning.

The EBLC path links each constraint field to its source, making mismatches traceable and available for review. A constraint can draw on an observed clearance time, a supplied waiting duration, and an assumption that the zone remains clear. A sentence alone may not identify which of these inputs is responsible for an error. EBLC separates them into source-linked fields that can be checked individually.

The composition diagnostic adds a distinct question: can conditions from separate source records hold together? Its three-way braking conflicts show that individual consistency, and even consistency of every pair, can leave a joint incompatibility undetected (Section 5.10). Identifying a subset-minimal source group makes the requirements needing review explicit. The source roles separate documented policy or guidance from authored observations, candidate requests, objectives, and modeling assumptions; their applicability is supplied rather than inferred. A SAT combination establishes compatibility under the specified context, while an UNSAT combination calls for reviewing requirements or assumptions rather than silently discarding one. Hold and non-crossing controls show why permission and progress obligations must be distinguished: waiting may satisfy restrictions even when a proposed movement is blocked. Comparing the faithful and omitted-exception Rule 129 encodings further identifies a lost permission, rather than a conflict in the original regulation; detecting an omitted exception therefore requires a reference encoding that retains it.

P1 is the natural-language guard path of our preceding work [21], and P2 is the EBLC path introduced here. The comparison therefore concerns two development paths for the same constraints. Both achieve high primary-test preferred-selection scores, and neither dominates every condition (Sections 5.4–5.7). P2 is ahead in position-hold continuation with XY coordinates under the 2.0 s rule (97.92% versus 91.15% preferred-selection score; 4 versus 16 violations of 192) and on pedestrian-entry paraphrases (96.94% versus 90.56%), whereas P1 is ahead on static paraphrases and the combined maneuver set. Because the paths also differ in wording, these differences are not attributed to specification checking. The EBLC path thus retains the behavioral benefit of constraint training and adds the ability to check constraint construction and correspondence.

Assurance in this process can be considered from three perspectives; the first two group the five checks in Section 3.5:

1. **Specification structure and logical correspondence.** Does the specification express the intended restriction and release within the supported model, rules, and queries? Structural validation, satisfiability checks, and logic–rule comparisons assess this question.
2. **Evidence and sentence correspondence.** Do specification fields match their source records, and does parsing the generated sentence with the fixed grammar recover those fields? Evidence and sentence correspondence checks assess these relationships.
3. **Model behavior.** Does the model select an admissible candidate, and does it select the preferred candidate under the stated objective? Constraint violation rate and preferred-selection score measure these two questions separately.

Fault injection shows complementary roles for the tested checks: specification-level checks detect faults in the specification and logical formula, whereas only sentence correspondence detects faults introduced in the generated CNL (Table 15). Evidence and sentence correspondence checks, together with boundary probes for logic–rule comparison, were added for this evaluation after the reported training runs.

Establishing the validity of a source record requires independent observations or rule evidence. All 48 controls in which the record and specification field are changed to the same wrong value remain internally consistent and escape every check. A possible division of work in practice would concentrate human review on the suitability of observations, rules, and assumptions, while automating structural, logical, and sentence-correspondence checks.

6.2 From Checked Meaning to Learned Interpretation

Specification checking and behavioral evaluation answer different questions: what the constraint says and how the model uses it. Recovering specification fields from a generated sentence does not establish how the model interprets that sentence. The results illustrate this distinction:

- **Paraphrasing.** Changing wording while retaining the intended rule reduces static-task P2 preferred-selection score from 100% to 80.32%; P1 reaches 93.29% under the same condition (Table 14).
- **Logical text.** Primary preferred-selection score is 99.17%, but on unseen waits it is 81.25% despite only 0.83% violations (Section 5.7).
- **Combined difficulty.** Combining negation, clause order, equality boundaries, and units reduces P2 rule-pair accuracy from 97.92% to 82.64% (Section 5.7).
- **Image dependence.** Although candidate values are supplied in text, a blank image reduces P2 preferred-selection score from 98.61% to 76.39% (Section 5.8). These interventions measure input dependence; visual grounding and event interpretation require separate assessment.

Multiple fixed expressions designed to recover the same specification fields offer a starting point for combining checkability with expression diversity. A fixed grammar enables sentence-correspondence checks, but a restricted set of training formulations may encourage dependence on particular wording. Whether training jointly with these expressions reduces wording sensitivity remains to be evaluated. Such checks apply within the defined grammar; they do not determine semantic equivalence for arbitrary natural-language sentences.

6.3 Learning Constraint-Conditioned Candidate Selection

Under the evaluated supervised-learning conditions, with initialization and evaluation inputs held fixed, training with constraint sentences improves rule-dependent candidate selection. In the common-CNL evaluation (Table 17), CE adapters trained with CNL reach 96.94% pedestrian-entry rule-pair accuracy, compared with 1.94% for CoC-only-trained adapters. This contrast concerns selecting both reference labels when the scene and candidates are identical but the rules differ: training with constraint text

improves the use of a rule available at evaluation relative to adapters fine-tuned in a rule-free format. Position-hold results show the same pattern.

Complementary evidence points in the same direction, although these evaluations do not isolate the training effect. Without task-specific training, the 8B model violates the supplied pedestrian-entry rule in 68.75% of predictions (Table 9). The preliminary Alpamayo-R1 trajectory diagnostics likewise exceed supplied speed limits in 15 of 18 outputs, with the same violation counts whether the rule is supplied through the user prompt or appended to CoC (Section 1; Supplementary Section S1). Together, these findings indicate that the presence of rule text at evaluation is insufficient for strong rule-dependent selection in the tested models, whereas training with constraint text in the evaluation format substantially improves that selection in the controlled study. This parallels instruction tuning, in which training with task instructions improves inference-time task performance [43]; here, the instruction is a case-specific constraint with numerical parameters, and its effect is tested by changing only the rule.

Providing constraints through the input and through the loss has different effects. CE supervises the reference label of the preferred candidate, while the semantic term in Eq. 9 increases the conditional probability mass assigned to admissible labels. Without rule text, the same scene and candidates can have different reference labels under two rules, leaving the inputs unable to distinguish the cases. Adding semantic loss nearly eliminates violations, but preferred-selection score remains near the rule-free ceilings of 50% and 75% (Table 10). This is consistent with more conservative choices admissible under both rules. Loss-based constraint learning encourages satisfaction [15, 45], whereas the input rule identifies which constraint applies. Adding the tested semantic term when rule text is already supplied gives similar results overall.

Reinforcement learning can also train a rule-conditioned policy when the applicable rule is available in its input. The present CE and semantic-loss comparisons establish the effect of constraint text within supervised fine-tuning; they do not establish superiority over reinforcement learning. Checking the supplied constraints remains a separate concern whichever optimizer is used, as discussed in Section 6.1.

Our learning objective is to select actions conditional on the rule supplied for each example. Constraints such as waiting durations or distance limits may differ even for the same scene because of local regulations, temporary signs, or operating instructions. The intended behavior is to receive the applicable constraint and adjust the choice accordingly. Deliberative Alignment trains models to recall reusable safety policies and apply them in context; policy text is used for data generation and filtering, then removed from supervised fine-tuning inputs [16]. Here, example-specific constraints and their numerical parameters remain in both training and evaluation inputs, and the evaluation tests whether choices respond to changes in those constraints.

6.4 Limitations and Future Work

Tasks and data. The learning experiments select from finite candidate sets in synthetic tasks. Candidate values are supplied in text, and static and maneuver diagrams are reused across splits. The results provide controlled evidence about constraint training; extending them to real-road visual understanding, trajectory generation,

or closed-loop safety requires separate evaluation. The Alpamayo-R1 diagnostics motivate the problem and do not constitute results from training Alpamayo.

Constraint compliance near boundaries. Selection improves on the new position-hold test, but the earlier XY regression diagnostic retains 106, 79, and 77 violations out of 192 predictions for P0, P1, and P2 under the 2 s rule after training (Section 5.3; Supplementary Section S11). This diagnostic includes prohibited candidates only 1 mm beyond the radius, compared with a minimum overshoot of 22 mm in the new test. The results therefore do not establish stable constraint compliance across conditions involving small overshoots.

Comparison design. Fine-tuning uses one 2B model family and three optimization seeds. Continuation comparisons include differences in prior training history; the fresh-adapter study separately controls initialization. CoC-only fine-tuning uses a rule-free input format, whereas common-CNL evaluation adds text absent during that fine-tuning. The comparison therefore includes adaptation to the constraint input format; it does not isolate how much of the gap arises from learning rule-conditioned selection versus reduced use of additional text after rule-free fine-tuning. The tested semantic loss acts only on conditional A–D label probabilities at the first supervised answer token, uses a weight fixed at $\lambda = 1$ before execution, and leaves alternative formulations to be compared. It provides a training penalty rather than a generation-time restriction or a vehicle-safety guarantee. The pretrained 8B evaluation includes task-format adaptation. No equivalence or noninferiority test between P1 and P2 was conducted. Their differences reflect wording and construction paths and do not isolate the causal effect of specification checking.

Checking. Because evidence correspondence, sentence correspondence, and boundary probes for logic–rule comparison were added after training, their effects on training-data quality and model behavior remain to be measured. Detection results concern supported grammars and injected fault types; the prevalence of naturally occurring errors must be estimated from actual constraint-authoring data. Evidence binding separates observations, rules, and assumptions by design, but the injection study did not test incorrect scene observations or unsuitable modeling assumptions; its faults concern rule values, logical restrictions, units, and generated CNL. The composition diagnostic uses a fixed, manually constructed grid grounded in selected document clauses. Authored states, simplified braking and signal models, candidate obligations, and one restricted exception define its scope. It does not evaluate complete regulations, emergency-trigger eligibility, jerk-limited braking, learned reward models, or operational safety. Its natural-language statements explain the formalization, rather than testing automatic legal interpretation or composition of CNL training examples. The accident examples motivate those broader checks; they are not evidence that this implementation would have prevented the accidents.

Future work should:

1. Evaluate how observations and rule applicability are assessed and bound in real scenes.
2. Apply constraint-bearing training data to trajectory-generating VLA models and measure trajectory-level constraint compliance, including near boundaries.

3. Evaluate deployment-time generation of constraint sentences from scene observations and applicable rules for provision to the model.
4. Test whether training with multiple expressions that recover the same specification fields reduces wording sensitivity.
5. Repeat the controlled fresh-adapter comparison with VLMs of other families and sizes.
6. Measure human review time and error rates for constraint authoring in an actual workflow.

7 Conclusion

Driving models need to use external behavioral rules as conditions for action selection. Constraint compliance and preference among admissible actions are separate requirements: a choice can satisfy the constraint while missing the preferred candidate under the stated objective. We proposed GuardSynth to make case-specific learning constraints explicit and checkable. It binds fields to observations, supplied rules, and modeling assumptions, represents the constraint in EBLC, checks satisfiability and candidate admissibility, and generates CNL for CoC training inputs with specification–sentence correspondence checks.

The behavioral evaluation measures constraint violations and selection of the preferred candidate separately. With matched initialization and identical CNL at evaluation, CE-only adapters trained with CNL achieve preferred-selection scores of 93.49–98.47%, constraint violation rates of at most 1.56%, and rule-pair accuracy of at least 86.98%. CoC-only-trained adapters reach 6.25–17.92% violations and at most 56.77% rule-pair accuracy. Neither semantic loss without constraint text nor the pretrained 8B model given the same rule reaches that paired performance. These findings support including case-specific constraints in supervised training as well as supplying them at inference for the evaluated adapters; they do not compare this approach with reinforcement learning.

Specification evaluation addresses the construction path separately. Fault injection into the two learning specifications detects all 288 supported faults and accepts all 48 valid cases. Specification-level and sentence checks detect different faults. All 48 jointly wrong evidence–specification controls remain undetected, showing the need for independent evidence confirmation. Evidence and sentence correspondence checks and boundary probes for logic–rule comparison were added after training, so their detection results are not attributed as the cause of behavioral improvement.

A separate bounded diagnostic combines documented policy and guidance with authored states, models, objectives, or candidate requests. It identifies 58 incompatible configurations among 176 cases, including 12 three-way conflicts whose pairs remain satisfiable, and retains 118 compatible configurations. This supports joint consistency checking under explicit contexts, separately from the learning results and general driving-safety claims.

The evidence comes from synthetic candidate selection, one fine-tuned 2B model family, and three optimization seeds. Rewording reduces preferred selection, and violations persist in a regression diagnostic with overshoots as small as 1 mm. Future

work should assess real-scene applicability, richer compositions of constraints and assumptions, trajectory-generating models, and deployment-time constraint provision. GuardSynth provides a development path that connects source-bound constraint construction, restricted specification checking, and language for learning while keeping constraint compliance and preferred selection as separate evaluation targets.

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Author contributions

Jin Hyun Kim conceived the study, conducted the experiments, and wrote the manuscript. Insup Lee contributed to the conception of the study and the evaluation of the results. Seung-Hyun Kong contributed to the evaluation of the research. Rahul Mangharam and Yon Vanommeslaeghe contributed to the evaluation of the results and review of the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare that they have no competing interests relevant to the content of this article.

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